

// **WAYSCONF 2026 · MASTERCLASS · 90 MINUTES**

Build an AI that checks and fixes its own work

Agentic loops in practice.

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// 0:00 – 0:08

The case

Most AI demos stop here

A prompt goes in. Something impressive comes out. Everyone claps.

Then someone asks a question the demo was not built to survive:

Does it match the design?

Is the contrast legal?

Does it work on a phone?

Is that the actual copy, or copy it invented?

no answer

This is a talk about the word *no*

A loop is only worth building if something in it can refuse
the work without asking a language model for an opinion.

What you will have at 16:25

A landing page for a music festival that does not exist, built from a Figma file by an agent you instructed, checked by a program that agent wrote and is not allowed to change, fixed when it failed, and deployed to a URL you can send to someone.

```
exit 0
```

YOU LEAVE WITH

A URL, live on the internet.

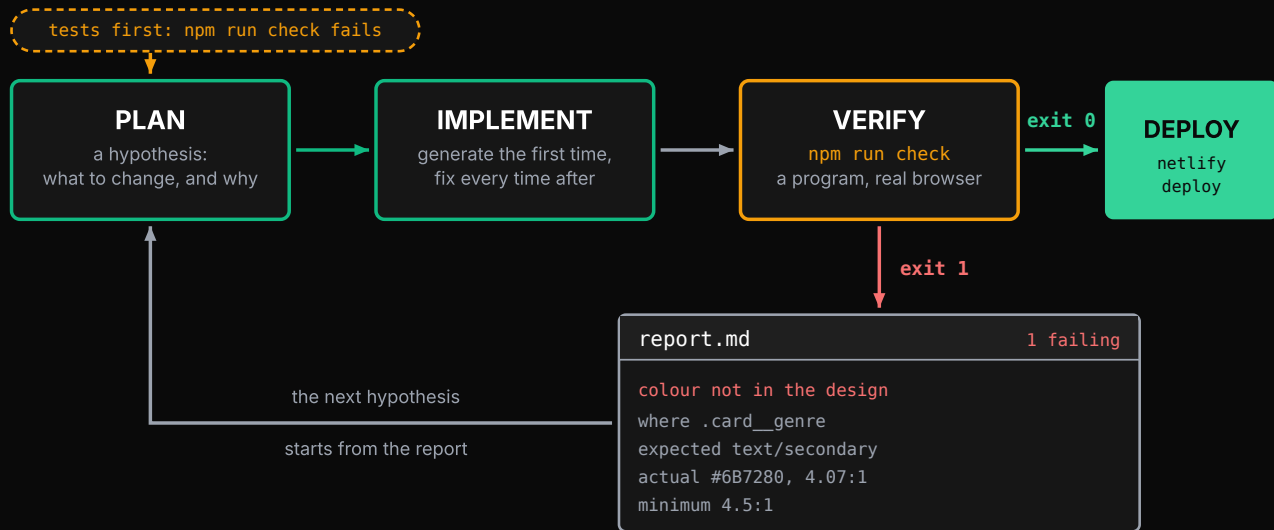
A folder: the page, and the checker that judged it.

Eight prompts that work on Monday, on something that is not a festival.

The loop

The loop

The check is written first and fails. Then plan, implement, verify, until it passes.



The way back carries a file. Nothing on it asks the model whether the work is good.

Setup

An agent running in an empty folder

The terminal, in one slide

```
pwd           # where am I
ls            # what is here
cd turbine    # go in there
claude        # or: codex
```



Esc stops the agent. **Ctrl+C** twice closes it:

```
claude --continue
```

 brings it back.

It is a text field that runs programs.

It gives you no feedback while it works, which reads as "broken" and is almost always "busy".

When a command finishes you get the prompt back. When you do not get the prompt back, it is still going.

Full version: the Preparation page, for your operating system

You need two things

THE DESIGN

`turbine.fig`, downloaded from the page — the file your agent reads. The same page opens the design in Figma, to look at.

THE PROMPTS

waysconf.szymonpaluch.com

Eight of them, with a copy button on each.

AND YOUR EMPTY REPOSITORY

```
cd turbine      # made at home
claude          # or: codex
```



It is **empty**, and there is no code to read.

Agent up? Paste **prompt 01** now.

Every prompt runs the same way

	WHAT YOU DO	CLAUDE CODE	CODEx
1. Plan	Switch to plan mode, paste the prompt, read the plan, approve it	<code>Shift+Tab</code> until <i>plan mode on</i>	<code>/plan</code>
2. Implement	The agent builds what the plan says		
3. Commit	The agent commits the step and pushes it to your repository	written into the prompt	written into the prompt
4. Clear	Empty the context before the next prompt	<code>/clear</code>	<code>/clear</code>

Prompts 02, 03 and 04 start in plan mode. The next prompt reads its state from files, not from the conversation.

Read the plan before it writes the code

Plan mode is the cheapest correction in the entire loop. A wrong assumption costs one sentence to fix here, and twenty minutes of generated code to fix later.

```
# Claude Code: Shift+Tab until the footer says "plan mode on"  
# Codex: /plan
```



Then read it. Actually read it. The plan is where you find out it misunderstood the brief.

// 0:15 — 0:25

Design in, not screenshots in

Plan mode, then prompt 02

Put `turbine.fig` in `design/` inside your project. Switch to **plan mode**, paste **prompt 02**, read the plan, approve it.

WHAT IT BUILDS

A decoder in your project: `npm run design`. It reads the `.fig` and writes the design as JSON in `design/data/`: sections, colours, typography, layout, components, copy, images.

The checker and the page both read those files. Nobody decodes or exports the design again.

WHAT YOU DO

Approve the plan, then wait ten to fifteen minutes. Let it use the network if it asks.

It writes no page. It stops with a list and its open questions (point 6). Answer them; it commits and pushes. Then `/clear`.

Two ways to hand over a design

A SCREENSHOT

The agent looks at pixels and infers.

The orange becomes *an* orange.

The spacing becomes *some* spacing.

It looks roughly right.

token gate fails

THE FILE ITSELF

Colours arrive as values. Spacing arrives as numbers.

```
TURBINE / Color
accent/sodium    #FF6A1A
text/secondary   #A7AEBB
text/muted       #6B7280
```



Nothing is guessed, so nothing is guessed wrong.

token gate passes

The file, and what happens to it

SO YOU HAND OVER THE FILE

turbine.fig	from the workshop page
canvas.fig	the whole document
images/	every photo



Nothing in this workshop is blocked by a free Figma account.

AND THE AGENT BUILDS A DECODER

It has never seen a `.fig`. It plans a decoder, writes it, runs `npm run design`, and fixes it until twelve cards carry twelve names. The result stays in the project as JSON, for the checker and for the page.

The verifier

Written before the page

Plan mode, then prompt 03

Answer 02's open questions first. Then `/clear`, switch to **plan mode**, paste **prompt 03**.

Your agent plans and writes `npm run check` from `design/data/`, before any page exists, and runs it on the empty project.

it must be red

About eight minutes. A check that passes on an empty page is not checking anything. It commits; then `/clear`.

Tests first

RED

The check is written from the design data and run on the empty project.

It fails. The report lists everything the page does not have yet.

prompt 03

GREEN

The page is built from the design data, then fixed against that report until the check passes.

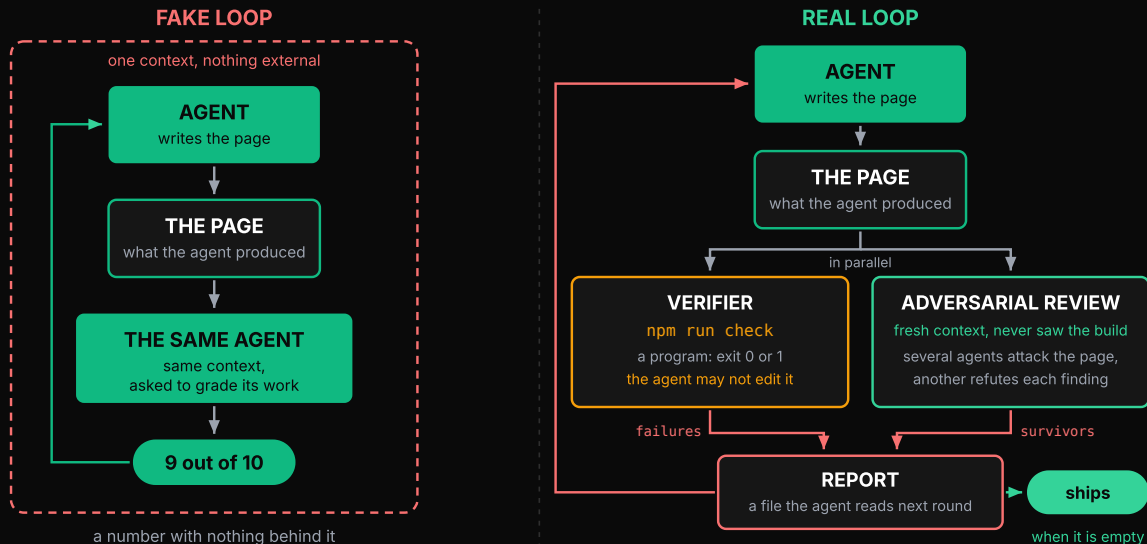
prompts 04 and 05

A check written after the page describes the page. A check written first can only describe the design.

A real loop and a fake one

Fake loop, real loop

An agent can check work. Not the same agent, in the same context, that did the work.



The program decides what can be measured. The fresh-context review looks for what cannot.
Neither replaces the other, and neither is the agent grading itself.

Who checks the work

WHO CHECKS

GOOD FOR

The same agent, same context

Nothing. It repeats its own reasoning

A program: `npm run check`

Anything with an exact answer: sections, colours, copy, contrast

Fresh-context agents, told to attack

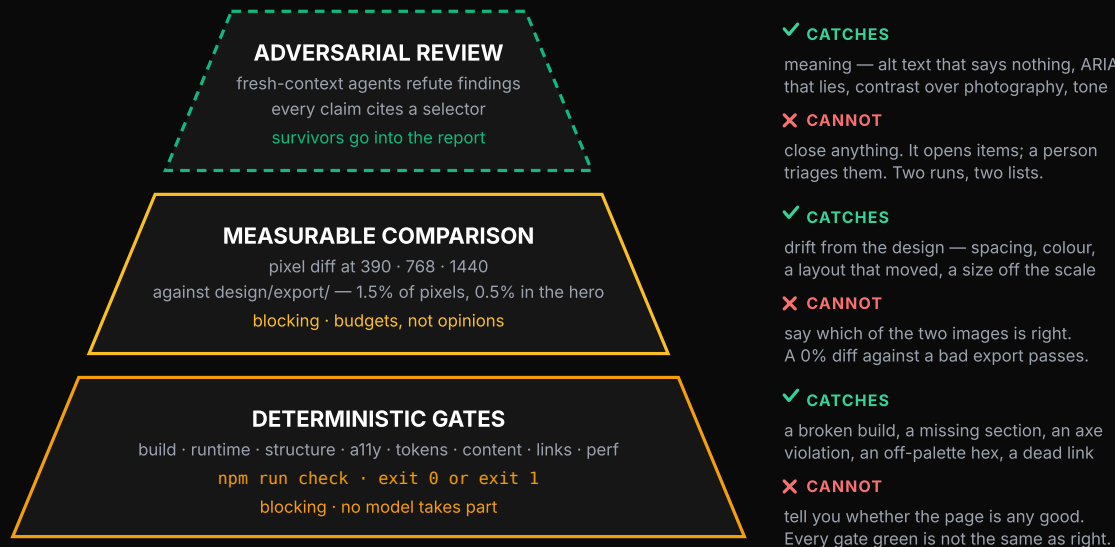
What a program cannot measure: reading order, alt text, text over photos, tone

Run the last two **side by side**. One report. It ships when the report is empty.

Three tiers of checking

Three tiers of verification

The bottom tier is the widest, the cheapest and the only kind that can be trusted to say no on its own.



The build gate runs first and alone; the seven browser gates then run together, and Lighthouse last.

Only the bottom two tiers can fail the build. Adversarial review runs alongside npm run check; what survives goes into the report.

Automated accessibility rules reach roughly 30 to 40 per cent of WCAG — a widely cited estimate, not a measurement of this page.

The reference site: ten gates, no opinions

GATE	WHAT IT DECIDES
Build	It compiles, no type errors, one page out
Runtime	No console errors, no failed requests
Structure	Sections, headings, landmarks, skip link, ids
Accessibility	axe clean at three widths, focus, keyboard, motion
Design tokens	Every colour and size came from the design
Visual fidelity	Pixels match the design, within a stated budget
Content	The real copy, all twelve artists, no placeholders
Links	Nothing points at nothing
Lighthouse	Performance, a11y, best practices, SEO, and five budgets
Text over images	The contrast axe refuses to judge

Watch it catch something real

The palette has a trap in it, on purpose.

```
color.text.muted    #6B7280  
color.bg.base       #0A0B0D
```



4.07 : 1

Under the 4.5:1 that body text requires.

```
color.text.secondary #A7AEBB  
color.bg.base        #0A0B0D
```



8.83 : 1

Ask any model for "a muted grey for supporting copy" and watch which one it reaches for.

The gate was green. The hero was illegible.

Zero axe violations. Three breakpoints. Six page states. Twice, deterministically. Lighthouse accessibility **100**.

Then I measured the hero by hand — screenshot the page with every glyph made transparent, sample the brightest patch behind each line, compute the real ratio.

TEXT	MEASURED
"Fourth edition"	2.94:1
"Ambient, techno and modular..."	3.02:1
"The Powerhouse, Hall E"	4.25:1
"Friday 12 – Sunday 14 June"	4.41:1

WHY AXE SAID NOTHING

It does not evaluate text over a background image. It does not fail it — it marks the pair **incomplete**, and in a zero-violations gate that is indistinguishable from correct.

Reading the CSS would not have found it either. The background there is a photograph, two scrims and a gradient composited together. No computed style says what colour that is.

That is why prompt 03 says: **measure text over images from the pixels, and fail if you cannot.**

The failure report is the interface

```
**3. [check 07] a colour on the page is not in the design**
```

- Where: the supporting line inside each card in the lineup section
- Expected: the design's "text / secondary"
- Actual: the design's "text / muted" – measured 4.07:1 against the page background,
and body text needs 4.5:1
- Hint: muted is defined for legal and footer text only



No model wrote that. A program measured it and printed it.

The agent's next iteration starts by reading this file.

The rule that holds it all up

The agent may never edit the verifier.

It is rule 1 of prompt 05, in capitals:

```
1. NEVER change the checker to make a check pass. Not a threshold,  
   not a skipped assertion, not a rule switched off. If you believe  
   a check is wrong, STOP, tell me which one and why.
```



Here is what actually happened

One agent. These prompts. An empty folder outside every repository, nothing else. Last night.

PROMPT	AGENT TIME	WHAT CAME OUT OF IT
01 — a project exists	3 min	Astro, dev server, first commit pushed
02 — reads the <code>.fig</code>	23 min	<code>npm run design</code> → seven JSON files, every colour and width
03 — the checker, before the page	20 min	eight checks, red against an empty project
04 — the page	24 min	ten sections in design order; two checks still red
05 — the loop	4 min	cleared thirteen accessibility failures, then stopped and asked
06 — it goes live	5 min	a URL, and the same check run against that URL

Seventy-nine minutes of agent time. Your ninety minutes are not that. **That is what the rescue packs are for**, and why they are on the clock, not on demand.

Minute four of the loop, it stopped

The checker wanted copy that the design has on **switched-off layers**. Two ways to make it green:

THE CHEAT

WHAT IT WOULD MEAN

`sr-only` text

A screen reader hears that the cheapest ticket is *"Most popular"*. It is not

`hidden` elements

Markup no user and no assistive tech ever reaches, added so a counter reads 3

It took neither. It wrote the reasoning into `notes.md`, named the check it believed was wrong, and asked.

It had already fixed thirteen real failures in the same four minutes. This is the one it would not fix.

And here is what went wrong

Thirteen incidents while building this. **Nine were failures of the verifier or the harness, not of the page.**

WHAT

Nine gates ran against **somebody else's website** for a whole run — 427 confident, correctly-formatted failures

axe passed a blank page. Zero violations, three breakpoints, green

The loop hung for an hour on a stdin nobody closed

The **evidence harness fabricated** four clean iterations out of a killed run

The gate was green and **the hero was illegible** — five of nine pieces of text below the legal minimum

WHY IT IS IN THE DECK

A verifier that is confident and wrong is worse than none

A measurement of nothing looks exactly like a measurement of perfection

Third unbounded wait in one project

Absence read as success — fourth time, and the only one that invented a table

axe does not fail text over a photograph. It marks it *incomplete*

Build the page

Your turn — 15 minutes

1. PLAN MODE, PROMPT 04

It plans the page from `design/data/`, builds all of it, and runs the check once. Still red, with fewer failures.

2. REVIEW IT

Beside Figma, one section at a time. Say what is wrong in plain words. Then it commits, and you `/clear`.

Falling behind cannot hurt you.

Stop where you are. Leave whatever is unfinished. I want to move on.

Prompts 02 to 05 build on each other. Behind at the rescue time? Take the pack from the page and go on.

// 0:55 — 1:10 • SPRINT 2 RUNS IN THE BACKGROUND

The loop, and workflows

Start your loop now, then listen

`/clear` , then paste **prompt 05**. Then leave it alone — I will talk over it, and when it stops it commits. If it is still red at the end of this block: Esc, then `/clear` and prompt 06.

```
Run npm run check. If it exits 0, stop and tell me.  
If it does not, read the report it wrote, fix what it names,  
run it again. Repeat until it exits 0.  
  
NEVER change the checker to make a check pass.
```



Shortened. Paste the full prompt 05 from the page.

Both agents also have a goal command:

```
/goal npm run check exits 0
```



The whole loop is four sentences

Check first	Run the check <i>before</i> you fix anything. An agent asked to improve passing work will find a reason.
Hand back the report	Unchanged. The file a program wrote, not your summary of it.
Never touch the checker	The moment the thing being judged can edit the judge, every green result afterwards means nothing.
Stop on a program	Not when it feels finished. When something exits 0.

No script. Nothing to install. You can say this to any agent, today.

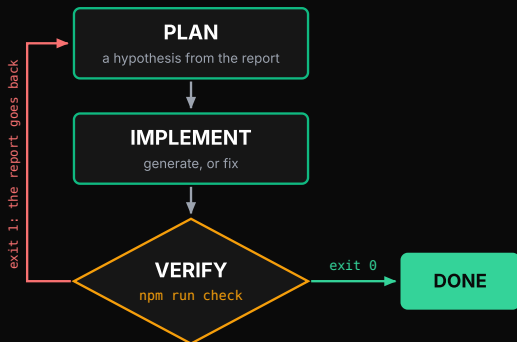
Two shapes, and neither is a script

Two shapes

A loop repeats until a condition holds. A workflow runs phases in order, and a phase can branch.

THE LOOP

repeat until a condition holds



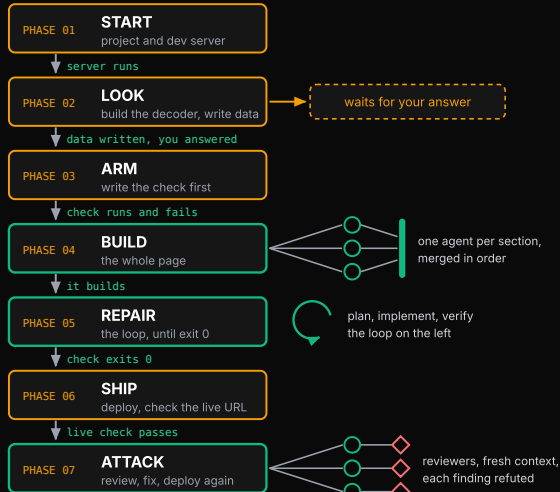
You define the exit condition.

Check written first. Then plan, implement, verify, until exit 0.

```
prompt 05 · /goal npm run check exits 0
```

THE WORKFLOW

phases in order, a bar between each



You define the phases, what branches inside them, and the bar between.

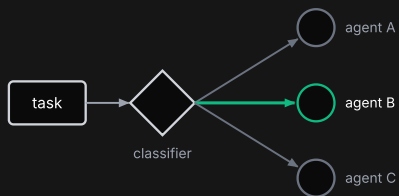
```
prompt 08
```

They nest: phase 05 of the workflow is the loop. Both are written as instructions, not code.

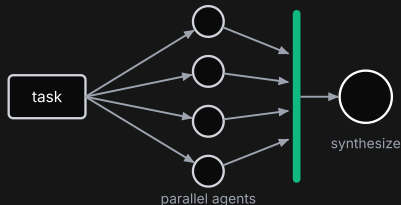
Six workflow patterns

Six workflow patterns

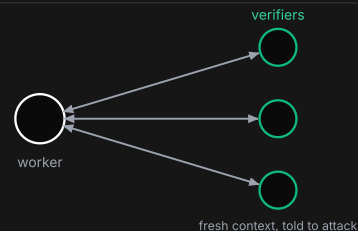
1 Classify and act



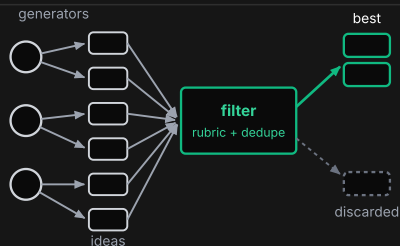
2 Fan out and synthesize



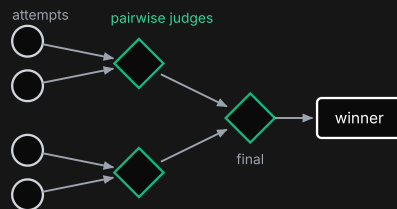
3 Adversarial verification



4 Generate and filter



5 Tournament



6 Loop until done



Why a fresh context beats a long one

ONE LONG CONVERSATION

Forty minutes of history.

Three abandoned approaches still in context.

The original brief, sixty thousand tokens ago.

Reasoning gets worse as it fills.

A FRESH CONTEXT EACH PASS

Reads the report — the current failures.

Reads `notes.md` — what was already tried, and what it cost.

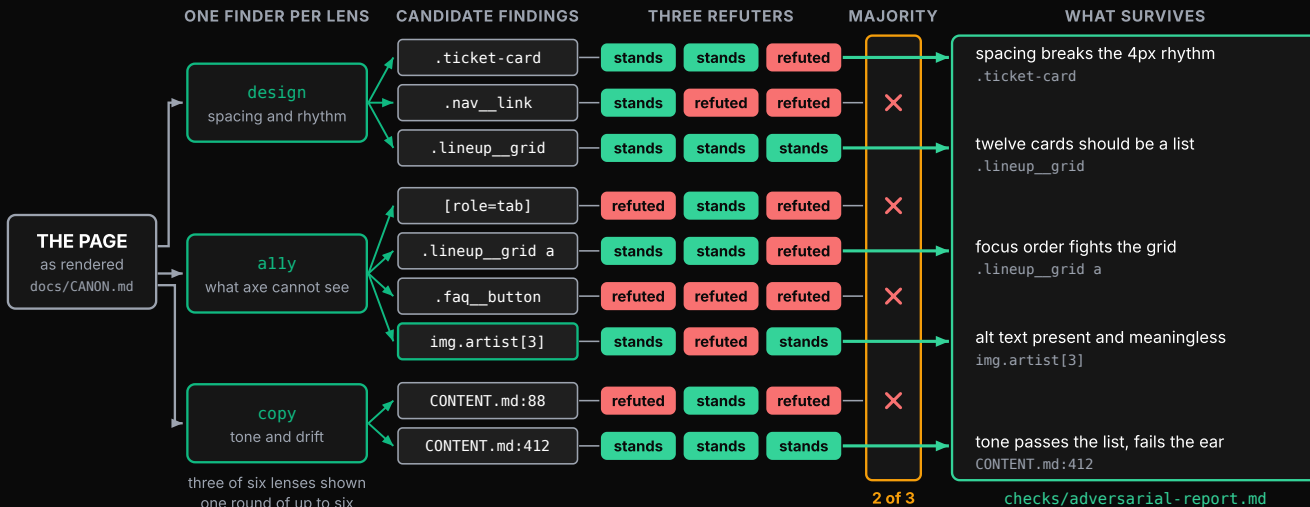
Reads design/data.

Notes on disk beat memory in context.

When one agent is not enough

One finder per lens, three refuters per finding

The work fans out, gets argued with, and only the part that survives a majority reaches a person.



Nothing here runs in lockstep.

Each candidate goes to three refuters whose only instruction is to argue that it is wrong, and reaches the filter when its own panel finishes. A lens that returns two findings does not hold up a lens that returns four.

The filter is arithmetic: two of three must stand. Every claim cites a selector or a file and line, or it is discarded unread.

Nothing on this page sets pass or fail. It opens items for a person to triage.

Adversarial review

The instruction is not "review this".

The instruction is **"refute this, and default to refuted if you are unsure"**.

```
ultracode. ... inside a phase, spawn as many subagents as the work  
needs and run them in parallel, deciding how many from what you find
```

```
PHASE 07 – ATTACK
```

```
... for every candidate they find, a separate subagent whose only  
job is to argue against it from those three angles.
```



That is **prompt 08**: prompts 01–07 as one message, phase for phase, with the differences it lists. Take it home. Today, **prompt 07** in a new session, after the deploy, is the one-agent version.

Ship, then break it

Deploy it, then try to break it

`/clear` , then **prompt 06**. It builds, creates your Netlify site without asking questions, deploys, and checks that the page the server returns is the page that passed. Take the URL it prints.

`/clear` , then **prompt 07**. An agent that never saw the build tries to break the page on your machine. Pick the findings that are real; it fixes them, the check still passes, and it deploys again.

Put your URL on the board. We are going to look at all of them.

What this does not do

Honesty

THE CLAIM

The a11y gate is green

The pixel diff passes

Adversarial review found nothing

```
npm run check
```

 exits 0

THE LIMIT

Automated rules reach maybe 30–40% of real WCAG failures

It saw no change. It has no idea whether a change would be better

A review finds what it looks for. It runs beside the program, never instead of it

Nothing here has an opinion about whether the design is good

Green means **no known defect**. It has never meant finished.

Where loops actually fail

It edits the test. The gate goes green and nothing was fixed. Forbid it in writing, and check that the file did not change.

It oscillates. Two wrong states, alternating. Usually two criteria that contradict each other. Cap the iterations and read the log.

It keeps going after it is done. Run the verifier *first*, every pass.

It runs out of context mid-task. Put the state in a file.

It reports success without running anything. Only believe the exit code.

It hangs. Not failing, just waiting. Every step needs a deadline — this project hung three times: six minutes, seven, and an hour.

Take this to work on Monday

Do not start with a whole landing page. Start with one gate.

1. Pick the check your team already does by hand and resents.
2. Make it a program that exits 0 or 1.
3. Put the failure output somewhere an agent can read.
4. Only then put an agent in front of it.

The loop is the easy part. The oracle is the work.

// EVERYTHING FROM TODAY

TURBINE

waysconf.szymonpaluch.com

The eight prompts, in English and Polish · the `.fig` · these slides
and the repository, with the checker and the evidence: `github.com/hculap/waysconf-2026-agentic-loops`

Questions. And then go and put your URL on the board.